

JS8Map

Installation & First Run

Version 1.75 · Windows 11 and Linux · Guide rev 1.55

Part of JS8MapChat

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*This guide covers JS8Map only. JS8FastChat has its own guide.
Read this once before you install anything — the order matters.*

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1 Before you start

JS8MapChat is the name for two programs that work together. There is no single program called JS8MapChat and no icon by that name.

JS8Map is a stand-alone app. Together with JS8FastChat it gives you a way to plot every JS8Call station on a map and then hold a fast chat with any of them, using JS8Call as the engine underneath.

Program	What it does	This guide?
JS8Map	Live map of the stations you are hearing and the stations hearing you.	Yes.
JS8FastChat	Operator console for directed messages, queries and group traffic.	No — it has its own guide. Optional; add it any time or never.
JS8Call	The digital mode software itself. JS8Map reads from it.	No. Required, and installed first.

JS8Map is a complete program on its own. It has its own installer, its own icon and its own window, and it does not need JS8FastChat to be installed at all.

THE ONE RULE ABOUT LAUNCH ORDER If you want to use JS8FastChat, launch JS8Map first. The FastChat button in the callsign popup depends on it. Install order and launch order are different things — install JS8Map first, and launch JS8Map first, every session.

1.1 Which version of JS8Call you are running

JS8Map works with both JS8Call 2.x and the newer JS8Call Improved 3.x. You do not need to upgrade JS8Call to use the map.

Some functions depend on features that exist only in the Improved 3.x version. On 2.x those controls are switched off rather than left to fail silently, so a greyed-out button is the program telling you your JS8Call cannot do that. It is not a fault in JS8Map.

Works on	What you get
Both 2.x and Improved 3.x	The map itself, station plotting, the activity console, tables, filters and searching. This is the bulk of the program.
Improved 3.x only	Halting transmission from within JS8Map. Changing speed from within JS8FastChat.

JS8Map carries a setting that tells it which JS8Call you have, and it enables or disables the affected controls to match. It is the TX Halt checkbox in the control window, labelled My JS8Call supports RIG.TX_HALT (Improved API 3.0+). Leave it unticked on 2.x. Set it after installing, once you know JS8Map is talking to JS8Call.

NOTE If you are on 2.x and need to stop a transmission, use JS8Call's own Halt TX button. It works exactly as it always has. Only the shortcut from inside JS8Map is unavailable.

1.2 What you need first

- JS8Call installed and working, with a callsign and grid square already set.
- JS8Call's TCP API turned on, listening on port 2442, and set to allow at least 5 connections. Section 1.3 covers this, and it is the single most important setting in this document.
- An internet connection, for the callsign databases JS8Map downloads during setup.
- Your Maidenhead grid square — for example FM05SX.

NOTE JS8Map will not connect until JS8Call is running with its API enabled. Installing with JS8Call closed is fine, but expect JS8Map to sit disconnected until you start it.

1.3 Turning on the JS8Call API — and the connection limit

In JS8Call, open File → Settings → Reporting and find the API section. Set it up exactly like this:

Setting	Set it to
Enable TCP Server API	Ticked.
Accept TCP Requests	Ticked. Without this, JS8Call ignores questions from JS8Map.
TCP Server Hostname	127.0.0.1
TCP Server Port	2442
TCP Max Connections	5 or more. The default of 1 is not enough — see below.

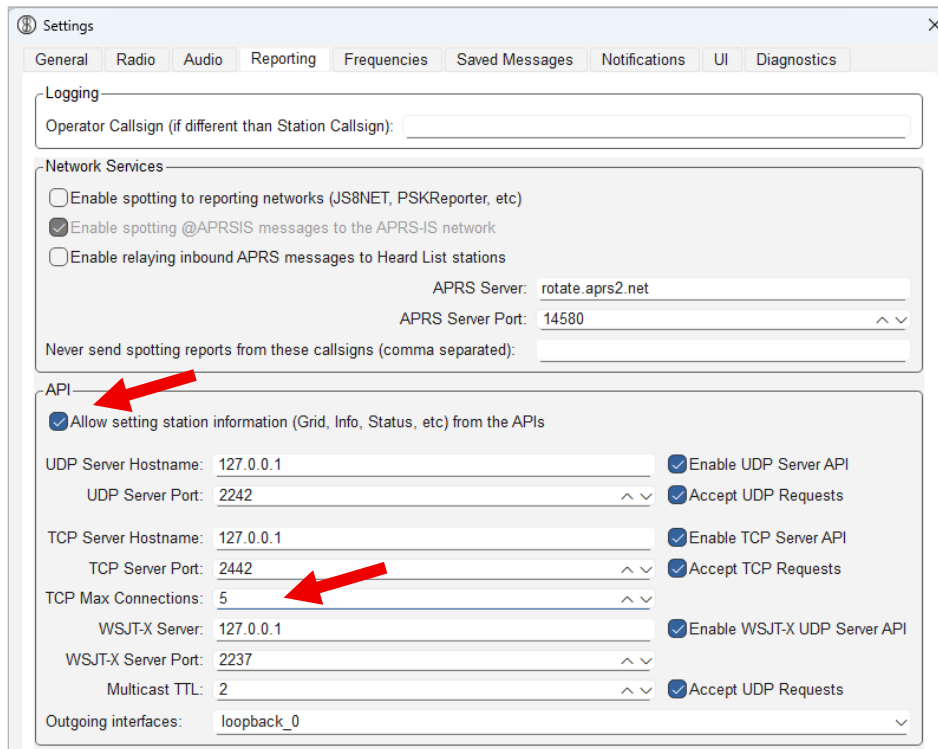


Figure 1 — The Reporting tab set up correctly. The API section is the lower half.

Click OK, then close JS8Call and reopen it. JS8Call only writes its settings to disk when it closes, so a change made without a restart leaves it running on the old setting.

READ THIS EVEN IF YOU SKIP THE REST TCP Max Connections is the number of programs allowed to talk to JS8Call at once, and it ships set to 1.

One connection is enough for JS8Map by itself. It stops being enough the moment you run anything else against JS8Call, and each of those programs needs a connection of its own. Count everything you run:

- JS8Map — one.
- JS8FastChat — one.
- JS8Spotter, CommStat Improved, or any other companion tool — one each.

At 1, the first program to start takes the only slot and everything after it is refused.

The refusal is silent. The program that lost still shows activity, because that comes from the shared database rather than from JS8Call, so it looks connected and healthy. What it cannot do is ask JS8Call anything — so frequency and offset boxes sit empty, commands do nothing, and no error is ever displayed. It looks like a fault in the program. It is not.

Set it to 5 and you have room for JS8Map, JS8FastChat and three other tools. Raise it further if you run more than that.

NOTE If JS8Map behaves perfectly and another program is oddly inert — or the reverse — check this setting before anything else. It is the most common cause and the least obvious.

2 Installing on Windows

You receive one file. It is an ordinary Windows installer — double-click it, answer a few questions, done.

File	What it installs
JS8Map_v1.75_Setup.exe	JS8Map, and everything it needs.

You do not need Python on Windows. The installer carries everything the program needs.

THE INSTALLER SAYS JS8MAPCHAT, NOT JS8MAP Its title bar reads Setup – JS8MapChat 1.75 and its last page says JS8MapChat 1.75 Installation Complete. That is the family name on the installer, and it is correct. This installer installs JS8Map. JS8FastChat has a separate installer of its own.

Accept the folder the installer offers, then leave both shortcut boxes ticked. The four pages look like this:

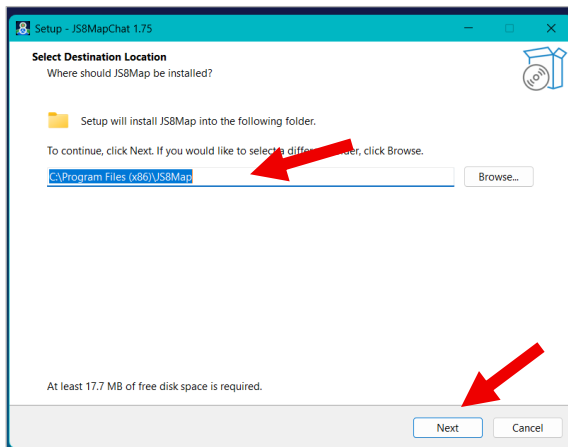


Figure 2 — Destination. Accept what is offered.

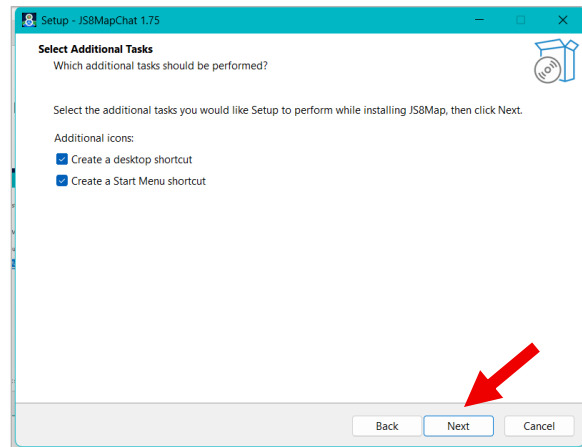


Figure 3 — Leave both shortcut boxes ticked.

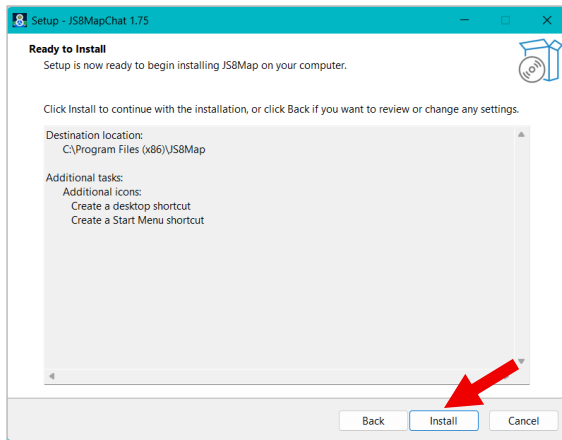


Figure 4 — Ready to install. Check the summary, then Install.

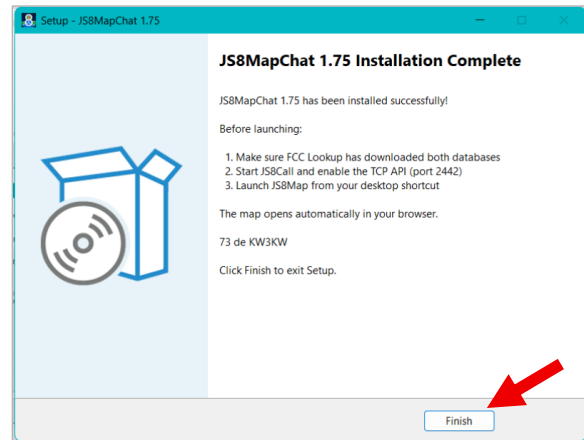


Figure 5 — Finished. The four steps it lists are covered in sections 1.3 and 5.

On a 64-bit Windows machine the program lands in C:\Program Files (x86)\JS8Map. That is expected and is not a sign that anything went wrong.

2.1 The desktop icon

When the installer finishes, a JS8Map icon appears on your desktop.

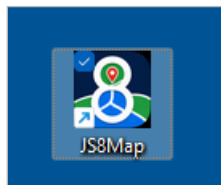


Figure 6 — The JS8Map desktop icon.

Click it to start JS8Map for the first time. That first launch is what activates the one-time setup wizard, which walks you through downloading the FCC and Canadian databases. Section 5 covers it step by step.

2.2 Updating an existing install

Windows upgrades install over the top. Do not uninstall first. Run the new installer and it finds the existing installation and replaces it in place.

IMPORTANT Accept the location the installer offers. Installing to a different folder leaves the old working install orphaned and running alongside the new one. Your settings and databases live outside the program folder and survive an upgrade either way.

2.3 If nothing seems to happen

On some machines the installer opens behind your other windows instead of in front, and all you see is a new button on the taskbar. Look there before double-clicking again, or press Alt+Tab. Starting it a second time just gives you two copies of the same installer.

The first launch of the program can also take up to a minute with nothing on screen, on a machine that has never run it before. Windows is checking the new file. It is not stuck. Wait before clicking again, for the same reason.

3 Installing on Linux

Linux has a real installer. You are given one compressed file, you unpack it, and you run one command. It installs into your own home folder, so there is no root, no sudo and no package manager involved.

YOU DO NOT NEED PYTHON The Linux package is self-contained. Python, tkinter and every library the program uses are inside it. Nothing needs installing with apt, dnf, pacman or pip. This is true even on a machine that has never had Python on it.

The one requirement is a reasonably modern distribution. The package is built to run on glibc 2.30 and newer, which covers Ubuntu 20.04 and anything more recent, and the equivalent releases of Mint, Debian, Fedora and Arch.

3.1 Installing

You receive one file, named JS8Map_v1.75_linux.tar.gz. **Save it in the Downloads folder.**

Now copy and paste this one command in Terminal:

```
cd ~/Downloads
tar -xzf JS8Map_v1.75_linux.tar.gz
cd JS8Map_v1.75_linux
./install.sh
```

IF IT SAYS PERMISSION DENIED A USB stick formatted FAT32 — which most are — cannot store the execute bit, so install.sh arrives unrunnable if you carried it over on one. Fix it with `chmod +x install.sh` and run the command again. Nothing is wrong with the package.

The installer checks that the package is complete before it copies anything, and it prints the contents of every folder it wrote to as its last step. Read that list. It is there so you can see the install landed where it should.

```
mj@mj-linux1: ~/Downloads/JS8Map_v1.75_linux
mj@mj-linux1:~$ cd ~/Downloads
tar -xzf JS8Map_v1.75_linux.tar.gz
cd JS8Map_v1.75_linux
./install.sh
=====
Installing JS8Map
=====

Installed to: /home/mj/.local/lib/JS8Map
An existing menu entry was saved as js8map.desktop.bak
An existing desktop icon was saved as JS8Map.desktop.bak

-----
Contents of /home/mj/.local/lib/JS8Map
-----
build_canadian_db.py
JS8Map
js8map_admin1.json
js8map_borders.json
js8map_lakes.json
js8map_places.json
js8map.png
leaflet.css
leaflet.js
-----

DONE. JS8Map is installed and all six map files are beside it.

Start it from the applications menu, from the desktop icon,
or by running: /home/mj/.local/lib/JS8Map/JS8Map

mj@mj-linux1:~/Downloads/JS8Map_v1.75_linux$
```

Figure 7 — A successful Linux install. The installer reports where it wrote the files, lists them, and tells you how to start the program.

NOTE install.sh refuses to run if it finds existing settings or database files where the program itself belongs. That refusal is protecting your data. If you see it, you are installing over something that is not a clean target — stop and check the path rather than forcing it.

When it finishes, a JS8Map icon appears on your desktop, along with a menu entry. It is the same icon shown in Figure 6. Click it to start the program for the first time, which activates the one-time setup wizard that downloads the FCC and Canadian databases. Section 5 covers it step by step.

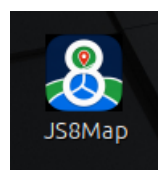


Figure 8 — The JS8Map desktop icon.

3.2 Updating an existing install

Download the new package and run it in the same place the current one is installed. It overwrites the outdated files and leaves the FCC and Canadian databases intact, so the wizard does not run again and you do not spend another twenty minutes downloading databases you already have.

WHY THE LOCATION MATTERS Installing somewhere new does not upgrade anything — it gives you a second, separate install running alongside the first. This is the same rule as on Windows, for the same reason.

3.3 If something goes wrong

Start the program from a terminal rather than the desktop icon. A desktop launcher swallows the error message; the terminal shows it. Run the command that `install.sh` printed, or simply type `JS8Map` if the installer put it on your path.

4 Where things are installed

The program itself is read-only. Everything that changes is kept somewhere else. That separation is why an update never loses your settings, and why uninstalling never loses your databases.

4.1 Windows

Location	What is kept there
C:\Program Files (x86)\JS8Map\	JS8Map and its map files. Nothing here needs editing.
%LOCALAPPDATA%\JS8Map\	Settings, the spot database, group tags and the generated map page.
%LOCALAPPDATA%\JS8MapChat\Shared\	The mailbox JS8Map and JS8FastChat pass messages through. Created whether or not you install JS8FastChat.
C:\FCC_DB\	The callsign databases. JS8Map creates and fills this folder, and JS8FastChat reads from it.

TIP To reach one of these folders, press Windows+R, type the path from the table and press Enter. You should not normally need to.

4.2 Linux

Location	What is kept there
~/.local/lib/JS8Map/	The program, its icon and its map data files. Nothing here needs editing.
~/.local/share/JS8Map/	Settings, the spot database, the map window profile and the generated map page.
~/.local/share/JS8MapChat/Shared/	The mailbox JS8Map and JS8FastChat pass messages through.
~/FCC_DB/	The callsign databases. JS8Map creates and fills this folder, and JS8FastChat reads from it.

NOTE An update replaces only the first of these four. The other three, including your callsign databases, are left alone deliberately — that is why an update does not send you back through the wizard.

5 First run — the setup wizard

Double-click the desktop icon. The first time JS8Map starts it walks you through a short setup: three steps, identical on Windows and Linux. The window is titled First-Time Setup and shows the three steps across the top, so you can always see where you are.

5.1 Step 1 — US callsign database

This is the large one. The download runs to roughly 190 MB straight from fcc.gov, and building the database from it takes a few minutes more. For the first half-minute it will look as though nothing is happening while the download starts. Let it run.

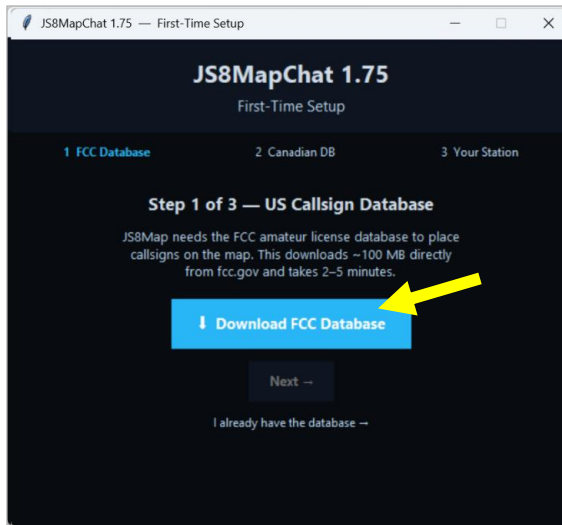


Figure 9 — Step 1, before you start. Click Download FCC Database.

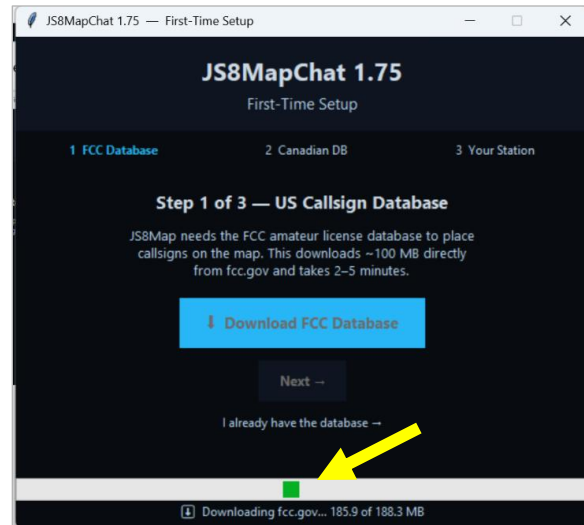


Figure 10 — Downloading. The progress line at the bottom shows the real size.

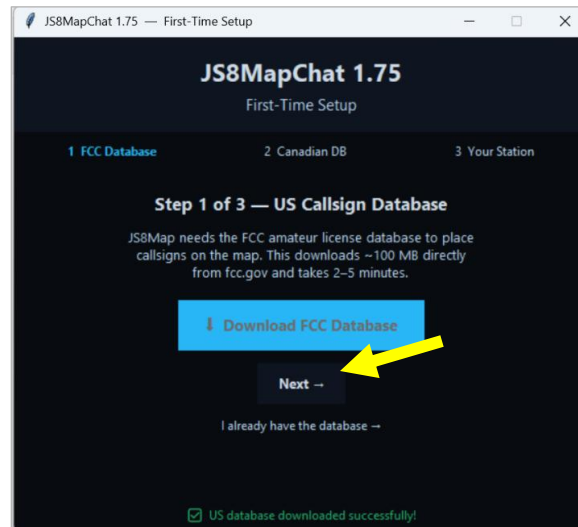


Figure 11 — Done. Next lights up once the database is built.

WHY STEP 1 HAS NO SKIP BUTTON Step 2 offers a Skip for now and step 1 does not, and that is deliberate. Canadian data is supplementary — the map works fully without it. The US database is not optional at all: without it JS8Map cannot place a single callsign on the map, because that is where the locations come from. Skipping it would only deliver you to a permanently blank map. If a download fails, the Download button is re-enabled so you can simply try again.

IF YOU ALREADY HAVE THE DATABASE The link under the buttons, I already have the database, is for a machine that has been through this before — a reinstall, or a second install pointed at an existing C:\FCC_DB folder. It is not a skip. Use it only if the database really is already on the machine.

5.2 Step 2 — Canadian callsign database

Adds VE, VA and VY coverage from ISED Canada. Much smaller, a minute or two, and genuinely optional — skip it and US stations still work normally. You can add it later from the main screen with the Update Canadian DB button.

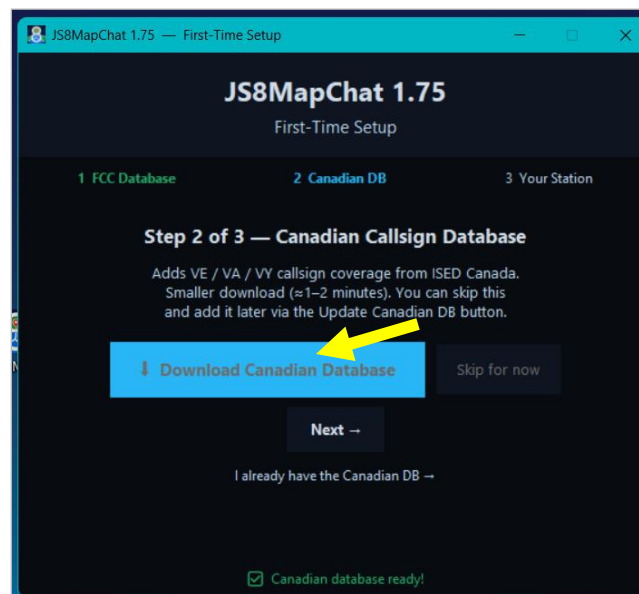


Figure 12 — Step 2. Skip for now is a real option here.

5.3 Step 3 — Your station

Callsign and Maidenhead grid, for example FM05SX. If JS8Call is running, this may fill itself in. You can change both later in Settings.

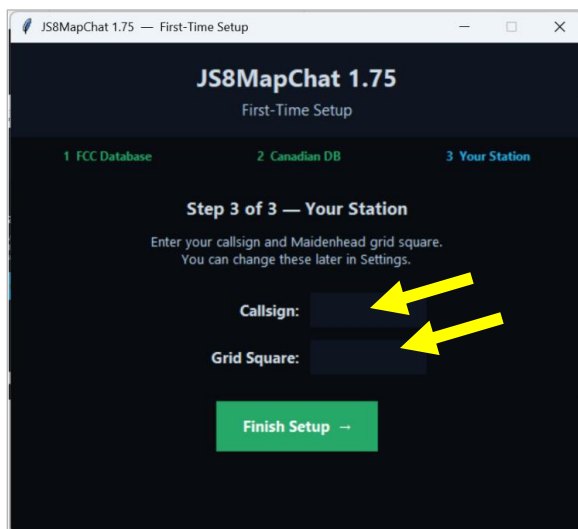


Figure 13 — Step 3, empty.

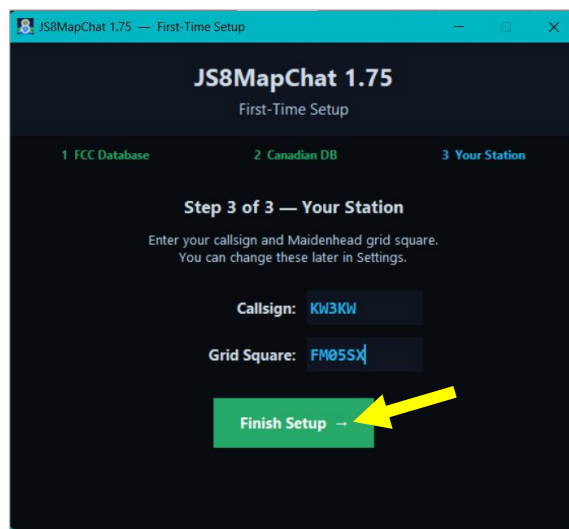


Figure 14 — Filled in. Click Finish Setup.

5.4 The control window

When setup finishes, the JS8Map control window opens. This window is not the map — it is the panel that talks to JS8Call and builds the map. The map itself opens in a window of its own when you click Generate Map.

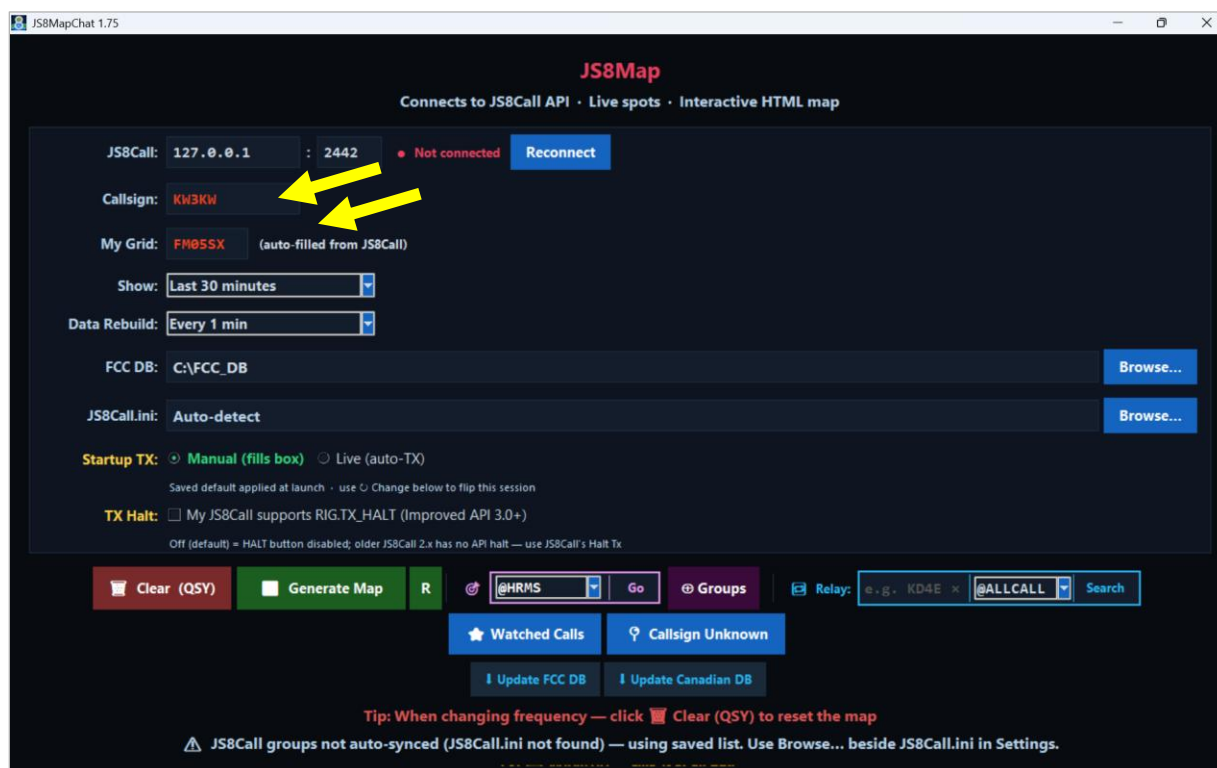


Figure 15 — The JS8Map control window. Connection state is top right; the buttons across the bottom are the ones you will use.

IF THE SETUP WINDOW DOES NOT APPEAR Check the taskbar. On some machines it opens behind whatever you were doing rather than in front. Alt+Tab brings it up.

5.5 Check the transmit mode before you connect

JS8Map can key your radio. Before connecting for the first time, look at Startup TX in the control window and decide which you want:

Setting	What happens
Manual (fills box)	Fills in the command for you to send from JS8Call. Nothing is transmitted on its own.
Live (auto-TX)	JS8Map transmits through JS8Call without asking again.

NOTE Manual is the safe choice while you are learning the program. This setting is the saved default applied at launch, and you can flip it for the current session at any time. Confirm which one you are in before you connect, so you never key the radio without meaning to.

6 How to tell it is working

Work down this list in order. Each line assumes the one above it passed.

What to look at	What a good result looks like
The connection indicator, top right of the control window	It reports a connection to JS8Call. If it reads Not connected in red, JS8Call is not running or its API is off — fix that, then click Reconnect.
Callsign and My Grid	Both filled in, with My Grid marked auto-filled from JS8Call. That is proof the API link is live, not just open.
The station count	It rises as JS8Call decodes stations. On a quiet band this can legitimately sit at zero for several minutes.
Click Generate Map	The map opens. Stations appear as markers. If you have heard nobody yet, an empty map is correct.
Click any station on the map	A popup opens with licence details, distance and signal relationship.
Pan the map east or west past the edge	The world repeats rather than running out. A station in Australia or Japan draws its line west across the Pacific, not east across Asia.

NOTE Give it time before deciding something is broken. JS8Map only knows what JS8Call has decoded, and JS8Call decodes on a fifteen-second cycle. A new install on a quiet evening can sit at zero stations and be entirely healthy.

7 When it does not work

These are the failures new installs actually hit, in rough order of how often they happen.

Symptom	Cause and fix
JS8Map does not connect to JS8Call. The indicator reads Not connected.	The TCP API is off, or JS8Call is not running. Turn it on in JS8Call's Reporting settings, confirm port 2442, then close and reopen JS8Call so the setting is written to disk. Then click Reconnect.
JS8Map looks connected and shows activity, but frequency and offset stay empty and commands do nothing.	TCP Max Connections in JS8Call is set to 1 and another program took the slot. Set it to 5, click OK, then close and reopen JS8Call. See section 1.3.
Nothing appears when you double-click the installer or the program.	Check the taskbar — it may have opened behind your other windows. A first launch can also take up to a minute on a machine that has not run it before.
The map generates, but no callsigns are placed.	The callsign database is missing or was never built. Click Update FCC DB on the control window and let it finish.
Canadian stations do not appear; US ones do.	The Canadian database was skipped during setup. Click Update Canadian DB. Nothing needs reinstalling.
A warning that JS8Call groups are not auto-synced and JS8Call.ini was not found.	JS8Map is using its saved group list instead of yours. Click Browse beside JS8Call.ini in the control window and point it at your JS8Call.ini.
The map is empty.	Often correct. Check the Show setting — a short window such as Last 30 minutes on a quiet band truthfully shows nothing. Widen it and see whether older stations reappear.
The map looks out of date.	Press R on the map to refresh it, or wait for the next Data Rebuild. If it is still stale, hard-refresh with Ctrl+F5.
A download fails at a different point each time you try.	That pattern means the remote server, not JS8Map. Try again later. A failure at the same point every single time is worth reporting.

8 Once you are running

8.1 Two map keys worth learning first

Both work on the map window itself, not the control window.

Key	What it does
F	Fits the map to your stations — resizes so every station is visible in one view. It doubles as an escape key: if you have zoomed or panned somewhere unhelpful, F returns the map to a sensible size.
R	Refreshes the map, pulling in the most current information without waiting for the next automatic rebuild.

8.2 Changing frequency — do it in this order

Nothing clears itself. When you change band, three things have to happen, in this order, across both programs:

1. Change frequency in JS8Call.
2. Click Clear List in JS8Call, to empty its decode list of the old band.
3. Click Clear (QSY) in JS8Map, to empty the map.

From that point the map shows only new decodes arriving on the new frequency, so every station on it is one you can actually work right now.

WHY STEP 2 IS NOT OPTIONAL If you clear the map but leave JS8Call's list full of the old band, those decodes are still there to be read and the map simply fills back up with stations you have left behind. Clearing JS8Call first is what makes clearing the map stick.

Stations on a band you have left cannot hear you and you cannot hear them, so leaving them on the map would be advertising contacts you cannot work. That is the whole reason for the sequence.

It follows that if you want to test how much history the map keeps, you must stay on one frequency for the whole test. Change bands partway through and you will clear the very history you were measuring.

8.3 Adding the operator console

If you want directed messages, queries and group traffic as well, that is JS8FastChat, and it has its own installation guide. Install it whenever you like. Remember the launch rule from section 1: start JS8Map first, then JS8FastChat.

The full manual covers every control in the program. You should not need it to get on the air.